

Row V, part 2b, the UV piece: every step, from your formula to the final answer

Part A explains each piece of mathematics that is used, with a proof or a numerical check. Part B applies them to your formula one step at a time. Nothing is skipped.

$$r = z - x, \quad s = y - x, \quad P = x' - w', \quad \xi = \frac{p^+}{k^+}, \quad \bar{\xi} = 1 - \xi, \quad q = \bar{\xi}k, \quad \lambda = \frac{\Lambda}{k^+}, \quad L = \log \frac{k^+}{\Lambda} = \log \frac{1}{\lambda}, \quad \gamma_E = 0.5772156649 \dots \text{ (Euler's constant)}$$

Where we will end up (derived in Steps 1–19):

$$\frac{d^3 N_{2b}^{\text{UV}}}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \mathcal{W} \left\{ \underbrace{\left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right]}_{\text{UV term}} + \underbrace{\frac{1 + e^{ik \cdot (y-x)}}{2} L^2 + c_0 L + H}_{\text{finite terms}} \right\} \mathcal{U}_b$$

- $\mathcal{N} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+}$ and $\mathcal{W} = \frac{(x'-w') \cdot (y-x)}{(x'-w')^2 (y-x)^2}$.
- $\mathcal{U}_b = [U^{ab'}(x') - U^{ab'}(w')] [-N_c U^{af}(x) + N_c U^{af}(y)] \rho^{b'}(x') \rho^f(y)$.
- c_0 and H are finite numbers, given in Steps 17 and 18.

Part A — the toolbox

A1. Vectors

All positions are 2-component transverse vectors. We write $r^2 = r \cdot r = |r|^2$. Expanding a square works as for numbers: $(r - s)^2 = r^2 - 2r \cdot s + s^2$.

A2. Polar coordinates, and what a "log divergence" is

Write $r = \rho(\cos \phi, \sin \phi)$, where $\rho = |r| \geq 0$ and $0 \leq \phi < 2\pi$. The area element is $d^2 r = \rho d\rho d\phi$. If q points along the first axis, $q \cdot r = |q|\rho \cos \phi$.

Example. Integrate $\frac{1}{r^2}$ over a small disc of radius a around $r = 0$:

$$\int_{|r| < a} \frac{d^2 r}{r^2} = \int_0^{2\pi} d\phi \int_0^a \frac{\rho d\rho}{\rho^2} = 2\pi \int_0^a \frac{d\rho}{\rho} = 2\pi [\log a - \log 0] = \infty.$$

- This is a **logarithmic UV divergence**: "UV" because it comes from short distances, $\rho \rightarrow 0$.
- By contrast, $\int \frac{d^2 r}{r} = 2\pi \int d\rho$ is finite near 0, and $\int_a^\infty \frac{d^2 r}{r^4} = 2\pi \int_a^\infty \frac{d\rho}{\rho^3}$ is finite at ∞ .

In d dimensions the same formula reads $d^d r = \rho^{d-1} d\rho d\Omega_d$, with total solid angle $\int d\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$. For $d = 2$ this gives $\frac{2\pi}{\Gamma(1)} = 2\pi$. (Γ is explained next.)

A3. The Gamma function

Definition: $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$. Its facts:

- $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$.
- $\Gamma(z+1) = z\Gamma(z)$. Proof by integration by parts: $\int_0^\infty t^z e^{-t} dt = [-t^z e^{-t}]_0^\infty + z \int_0^\infty t^{z-1} e^{-t} dt$, and the bracket is 0. Applied at $z = -\epsilon$, it gives $\Gamma(-\epsilon) = \frac{\Gamma(1-\epsilon)}{-\epsilon}$.
- **Near 1:** $\Gamma(1+x) = 1 - \gamma_E x + O(x^2)$ for small x . (This is in fact the definition of γ_E : $\Gamma'(1) = -\gamma_E$.)
- **Beta integral**, for $0 < a < 1$:

$$\int_0^\infty \frac{t^{a-1}}{1+t} dt = \Gamma(a) \Gamma(1-a) = \frac{\pi}{\sin \pi a}. \quad (\text{A3})$$

Numerically at $a = 0.2, 0.5, 0.8$: 5.3447966606, 3.1415926536, 5.3447966606 on both sides.

- **Powers:** $c^{-\epsilon} = e^{-\epsilon \log c} = 1 - \epsilon \log c + O(\epsilon^2)$.

A4. Bessel functions J_0 , K_0 , K_1

These are standard functions (like sin or log), tabulated and built into every math library. We only need the following facts.

- J_0 is what the angular integral gives:

$$\int_0^{2\pi} d\phi e^{iz \cos \phi} = 2\pi J_0(z), \quad J_0(0) = 1.$$

Check: at $z = 0.3$ and $z = 2$, 6.1426068682 and 1.4067472539 on both sides.

- K_0 is the 2D Fourier transform of $\frac{1}{u^2+m^2}$ (Gradshteyn–Ryzhik 6.532.4). Do the angle with the line above, then $\int_0^\infty \frac{u J_0(qu)}{u^2+m^2} du = K_0(qm)$, so

$$\int d^2u \frac{e^{iq \cdot u}}{u^2 + m^2} = 2\pi K_0(|q|m). \quad (\text{A4})$$

Check: 0.89211600 vs 0.89211600 at $(q, m) = (1.3, 0.4)$; 0.42102444 vs 0.42102444 at $(0.5, 2)$.

- **Small argument:** $K_0(z) = -\log \frac{z}{2} - \gamma_E + O(z^2 \log z)$. Check at $z = 10^{-5}$: 11.62885698 both.
- **Large argument:** $K_0(z) \approx \sqrt{\frac{\pi}{2z}} e^{-z}$, which is tiny.
- K_1 is defined by $K_1(z) = -\frac{dK_0}{dz}$. Near 0, $K_1(z) \approx \frac{1}{z}$, and $0 < z K_1(z) \leq 1$ for all $z > 0$.

A5. Moving a derivative inside a Fourier integral

If $f(u) \rightarrow 0$ at infinity, integration by parts gives

$$\int d^2u e^{iq \cdot u} \frac{\partial f}{\partial u^j} = - \int d^2u \frac{\partial e^{iq \cdot u}}{\partial u^j} f = -iq^j \int d^2u e^{iq \cdot u} f.$$

Also, derivatives with respect to a parameter pass through the integral: $\frac{\partial}{\partial \Delta} \frac{1}{u^2 + \Delta} = -\frac{1}{(u^2 + \Delta)^2}$.

A6. Feynman's trick for a product of two denominators

$$\frac{1}{ab} = \int_0^1 \frac{dt}{[ta + (1-t)b]^2}.$$

Proof. The denominator is $[b + t(a-b)]^2$, so

$$\int_0^1 \frac{dt}{[b + t(a-b)]^2} = \left[\frac{-1}{(a-b)(b + t(a-b))} \right]_0^1 = \frac{-1}{(a-b)a} + \frac{1}{(a-b)b} = \frac{a-b}{(a-b)ab} = \frac{1}{ab}.$$

(Check with $a = 2.3$, $b = 0.6$: 0.72463768 both.)

A7. Dimensional regularisation and $\overline{\text{MS}}$

The problem. $\int \frac{d^2r}{r^2}$ is infinite (A2). **The trick:** do it in $d = 2 - 2\epsilon$ dimensions. Near 0 the integral becomes $\int \rho^{d-1} \frac{d\rho}{\rho^2} = \int \rho^{-1-2\epsilon} d\rho$, which is finite if $\epsilon < 0$. The answer then has a pole $\propto \frac{1}{\epsilon}$, and the pole is the divergence. The conventions:

- Multiply by $\mu^{-2\epsilon}$ (μ has units of momentum), so that the result keeps the units it had in $d = 2$.
- Write the pole as $\frac{1}{\epsilon_{\text{UV}}} \equiv -\frac{1}{\epsilon}$. It is positive, since the integral converged only for $\epsilon < 0$.
- $\overline{\text{MS}}$ scale: $\mu_{\overline{\text{MS}}}^2 \equiv 4\pi e^{-\gamma_E} \mu^2$. Equivalently, $\pi \mu^2 = \frac{\mu_{\overline{\text{MS}}}^2 e^{\gamma_E}}{4}$. This absorbs the 4π and γ_E that always accompany the pole.

Anything finite can be computed directly at $d = 2$: its ϵ corrections vanish as $\epsilon \rightarrow 0$.

A8. The plus prescription

Suppose $f(\xi)$ is finite at $\xi = 0$ and we need $\int_\lambda^1 \frac{f(\xi)}{\xi} d\xi$ as $\lambda \rightarrow 0$. Subtract and add $f(0)$:

$$\int_\lambda^1 \frac{f(\xi)}{\xi} d\xi = \int_\lambda^1 \frac{f(\xi) - f(0)}{\xi} d\xi + f(0) \int_\lambda^1 \frac{d\xi}{\xi} = \underbrace{\int_0^1 \frac{f(\xi) - f(0)}{\xi} d\xi}_{\text{finite}} + f(0) L + O(\lambda).$$

- The first integral is finite because $f(\xi) - f(0) \approx f'(0) \xi$ cancels the $\frac{1}{\xi}$.
- The second is $\int_\lambda^1 \frac{d\xi}{\xi} = \log \frac{1}{\lambda} = L$.

- Written as a distribution: $\frac{1}{\xi} = \left[\frac{1}{\xi}\right]_+ + \delta(\xi) L$.

With a log. $\int_{\lambda}^1 \frac{\log(1/\xi)}{\xi} d\xi = \left[-\frac{1}{2} \log^2 \xi\right]_{\lambda}^1 = \frac{1}{2} L^2$. So if $f(\xi) \approx A \log \frac{1}{\xi} + B$ near 0:

$$\int_{\lambda}^1 \frac{f(\xi)}{\xi} d\xi = \int_0^1 \frac{f(\xi) - A \log \frac{1}{\xi} - B}{\xi} d\xi + \frac{A}{2} L^2 + B L + O(\lambda \log^2 \lambda). \quad (\text{A8})$$

The same holds at $\xi \rightarrow 1$ with $\bar{\xi} = 1 - \xi$ in place of ξ . (Check with $f = \cos \xi + 2\xi$ at $\lambda = 10^{-6}$: 15.5756968 vs 15.5756988; and $\int_{\lambda}^1 \frac{\log(1/\xi)}{\xi} = 95.43416599 = \frac{1}{2} L^2$.)

Part B — the calculation

Step 1. Collect the constants

$$\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \times \frac{1}{2\pi} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} = \frac{\mathcal{N}}{\pi}, \quad \mathcal{N} \equiv \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+}.$$

Step 2. See what depends on what

factor	depends on
$e^{-ik \cdot w'} e^{ik \cdot x}$	x, w' only
colour $\mathcal{U}_b = [U^{ab'}(x') - U^{ab'}(w')] [-N_c U^{af}(x) + N_c U^{af}(y)] \rho^{b'}(x') \rho^f(y)$	x, y, x', w' only
$C_{UV}(\xi) \equiv \xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\bar{\xi}}$	ξ only
$e^{iq \cdot r}, \frac{1}{\xi D}, \frac{1}{r^2}$	r and ξ (and s, k)
$\frac{s \cdot P}{P^2}$	s, P only

So the colour factor, the phase and $\frac{s \cdot P}{P^2}$ can be taken outside the r and ξ integrals.

Step 3. Separate the P dependence

$$\frac{s \cdot P}{P^2 r^2} = \frac{s \cdot P}{P^2 s^2} \cdot \frac{s^2}{r^2} = \mathcal{W} \frac{s^2}{r^2}, \quad \mathcal{W} \equiv \frac{s \cdot P}{P^2 s^2}.$$

Step 4. The formula after Steps 1–3

$$\frac{d^3 N_{2b}^{UV}}{d^3 k} = \mathcal{N} \int_{x, y, x', w'} e^{-ik \cdot w'} e^{ik \cdot x} \mathcal{W} \mathcal{U}_b \int_{\lambda}^{1-\lambda} d\xi C_{UV}(\xi) \frac{K(\xi)}{\pi}, \quad K(\xi) \equiv \int d^2 r e^{iq \cdot r} \frac{s^2}{\xi D r^2}.$$

All the work is now in the single integral $K(\xi)$.

Step 5. What D is

$D = (r - s)^2 + M^2$ with $M^2 = \frac{\xi}{\bar{\xi}} s^2$. At $r = 0$: $D_0 = s^2 + \frac{\xi}{\bar{\xi}} s^2 = \frac{\xi + \bar{\xi}}{\bar{\xi}} s^2 = \frac{s^2}{\bar{\xi}}$, as you wrote. Two facts used below: $\bar{\xi} D_0 = s^2$, and $M^2 = D_0 - s^2$.

Step 6. Where is K infinite?

Check the three places where an integral can blow up (A2):

- $r \rightarrow 0$. $D \rightarrow D_0$, so $\frac{s^2}{\xi D} \rightarrow \frac{s^2}{\xi D_0} = 1$, and $e^{iq \cdot r} \rightarrow 1$. The integrand becomes $\frac{1}{r^2}$: **logarithmically divergent. This is the UV divergence.**
- $r \rightarrow s$. $\frac{1}{D} = \frac{1}{(r-s)^2 + M^2} \leq \frac{1}{M^2}$: finite for $\xi > 0$.
- $|r| \rightarrow \infty$. $D \approx r^2$, so the integrand $\approx \frac{s^2}{\xi r^4}$, and $\int \frac{d\rho}{\rho^3}$ is finite.

So there is exactly one divergence, at $z \rightarrow x$. Everything else is finite.

Step 7. Split the integrand exactly into three simpler pieces

Expand D with A1: $D = r^2 - 2r \cdot s + s^2 + M^2 = r^2 - 2r \cdot s + D_0$. Rearranged:

$$D_0 = D + 2r \cdot s - r^2.$$

Since $\frac{s^2}{\xi} = D_0$:

$$\frac{s^2}{\xi D r^2} = \frac{D_0}{D r^2} = \frac{D + 2r \cdot s - r^2}{D r^2} = \frac{1}{r^2} + \frac{2r \cdot s}{r^2 D} - \frac{1}{D}.$$

Multiply by $e^{iq \cdot r}$ and integrate:

$$K = K_I + K_{II} - K_{III}, \quad K_I = \int d^2 r \frac{e^{iq \cdot r}}{r^2}, \quad K_{II} = \int d^2 r e^{iq \cdot r} \frac{2r \cdot s}{r^2 D}, \quad K_{III} = \int d^2 r \frac{e^{iq \cdot r}}{D}.$$

- K_I has the whole divergence ($\frac{1}{r^2}$ at 0).
- K_{II} is finite: at 0, $\frac{r \cdot s}{r^2} \sim \frac{1}{\rho}$; at ∞ it goes as $\frac{1}{\rho^3}$.
- K_{III} is finite: $D \geq M^2$, and the phase tames the $\frac{1}{\rho^2}$ tail (see Step 11).

Step 8. Split K_I into a UV part and a finite part

$$\frac{e^{iq \cdot r}}{r^2} = \underbrace{\frac{s^2}{r^2(r^2 + s^2)}}_{\text{UV part}} + \underbrace{\frac{1}{r^2} \left[e^{iq \cdot r} - \frac{s^2}{r^2 + s^2} \right]}_{\text{finite part}} \quad (\text{the two } \frac{s^2}{r^2(r^2 + s^2)} \text{ cancel}).$$

Why this choice:

- Near $r = 0$ the UV part is $\approx \frac{1}{r^2}$, the same as the original, so it carries the whole divergence.
- It has no phase and falls like $\frac{1}{r^4}$. Its only scale is $s^2 = (y - x)^2$, so the log comes out in transverse coordinates.
- In the finite part, the bracket $\approx (1 + iq \cdot r) - (1 - \frac{r^2}{s^2}) \approx iq \cdot r$, so it behaves like $\frac{1}{\rho}$ at 0: finite.

So $K_I = K_I^{\text{UV}} + K_I^{\text{fin}}$.

Step 9. K_I^{UV} in dimensional regularisation (A7)

(a) Replace $d^2 r \rightarrow \mu^{-2\epsilon} d^d r$ with $d = 2 - 2\epsilon$: $K_I^{\text{UV}} = \mu^{-2\epsilon} \int d^d r \frac{s^2}{r^2(r^2 + s^2)}$.

(b) The integrand depends only on $\rho = |r|$, so the angles give the total solid angle (A2), $\frac{2\pi^{d/2}}{\Gamma(d/2)} = \frac{2\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}$:

$$K_I^{\text{UV}} = \mu^{-2\epsilon} \frac{2\pi^{1-\epsilon}}{\Gamma(1-\epsilon)} \int_0^\infty \rho^{1-2\epsilon} d\rho \frac{s^2}{\rho^2(\rho^2 + s^2)}.$$

(c) Put $u = \rho^2$. Then $du = 2\rho d\rho$, so $\rho^{1-2\epsilon} d\rho = \frac{1}{2} u^{-\epsilon} du$:

$$\int_0^\infty \rho^{1-2\epsilon} d\rho \frac{s^2}{\rho^2(\rho^2 + s^2)} = \frac{s^2}{2} \int_0^\infty du \frac{u^{-\epsilon-1}}{u + s^2}.$$

(d) Put $u = s^2 t$. Then $du = s^2 dt$, $u^{-\epsilon-1} = (s^2)^{-\epsilon-1} t^{-\epsilon-1}$ and $u + s^2 = s^2(1 + t)$:

$$\int_0^\infty du \frac{u^{-\epsilon-1}}{u + s^2} = (s^2)^{-\epsilon-1} \int_0^\infty \frac{t^{-\epsilon-1}}{1 + t} dt = (s^2)^{-\epsilon-1} \Gamma(-\epsilon) \Gamma(1 + \epsilon).$$

The last equality is the Beta integral (A3) with $a = -\epsilon$. It needs $0 < -\epsilon < 1$, i.e. $\epsilon < 0$, as A7 said.

(e) Multiply (b), (c) and (d): $\frac{2\pi^{1-\epsilon}}{\Gamma(1-\epsilon)} \cdot \frac{s^2}{2} \cdot (s^2)^{-\epsilon-1} = \pi \frac{\pi^{-\epsilon}(s^2)^{-\epsilon}}{\Gamma(1-\epsilon)}$. Together with $\mu^{-2\epsilon}$:

$$K_I^{\text{UV}} = \pi (\pi \mu^2 s^2)^{-\epsilon} \frac{\Gamma(-\epsilon) \Gamma(1 + \epsilon)}{\Gamma(1 - \epsilon)}.$$

(f) Simplify with $\Gamma(-\epsilon) = \frac{\Gamma(1-\epsilon)}{-\epsilon}$ (A3). The $\Gamma(1 - \epsilon)$ cancels, and then expand $\Gamma(1 + \epsilon)$:

$$\frac{\Gamma(-\epsilon) \Gamma(1 + \epsilon)}{\Gamma(1 - \epsilon)} = -\frac{\Gamma(1 + \epsilon)}{\epsilon} = -\frac{1 - \gamma_E \epsilon + O(\epsilon^2)}{\epsilon} = -\frac{1}{\epsilon} + \gamma_E + O(\epsilon).$$

(g) Expand the power (A3) and multiply. Only $\frac{1}{\epsilon} \times \epsilon$ terms give extra constants:

$$(1 - \epsilon \log(\pi \mu^2 s^2)) \left(-\frac{1}{\epsilon} + \gamma_E \right) = -\frac{1}{\epsilon} + \gamma_E + \log(\pi \mu^2 s^2) + O(\epsilon).$$

(h) $\overline{\text{MS}}$ (A7): $\pi \mu^2 = \frac{\mu_{\overline{\text{MS}}}^2 e^{\gamma_E}}{4}$, so $\gamma_E + \log(\pi \mu^2 s^2) = \log \frac{\mu_{\overline{\text{MS}}}^2 s^2 e^{2\gamma_E}}{4}$. With $-\frac{1}{\epsilon} = \frac{1}{\epsilon_{\text{UV}}}$:

$$K_I^{\text{UV}} = \pi \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\overline{\text{MS}}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right].$$

Step 10. K_I^{fin} (finite, so done at $d = 2$)

(a) Polar coordinates with q along the first axis (A2). The angular integral is (A4): $\int_0^{2\pi} d\phi [e^{i|q|\rho \cos \phi} - \frac{s^2}{\rho^2 + s^2}] = 2\pi [J_0(|q|\rho) - \frac{s^2}{\rho^2 + s^2}]$. Therefore

$$K_I^{\text{fin}} = \int_0^\infty \rho d\rho \frac{1}{\rho^2} 2\pi \left[J_0(|q|\rho) - \frac{s^2}{\rho^2 + s^2} \right] = 2\pi \int_0^\infty \frac{d\rho}{\rho} \left[J_0(|q|\rho) - \frac{s^2}{\rho^2 + s^2} \right].$$

(b) Put $x = \frac{\rho}{|s|}$. Then $\frac{d\rho}{\rho} = \frac{dx}{x}$, $\frac{s^2}{\rho^2 + s^2} = \frac{1}{1+x^2}$ and $|q|\rho = ax$ with $a \equiv |q||s|$:

$$K_I^{\text{fin}} = 2\pi R(a), \quad R(a) \equiv \int_0^\infty \frac{dx}{x} \left[J_0(ax) - \frac{1}{1+x^2} \right].$$

(c) **Computing $R(a)$.** The two terms cannot be integrated separately, because each diverges at $x = 0$. Insert a factor x^η with small $\eta > 0$, which makes each term finite on its own. Since the combination is already finite, letting $\eta \rightarrow 0$ at the end gives $R(a)$. The two pieces are:

- $\int_0^\infty x^{\eta-1} J_0(ax) dx = 2^{\eta-1} a^{-\eta} \frac{\Gamma(\eta/2)}{\Gamma(1-\eta/2)}$ (Gradshteyn–Ryzhik 6.561.14).
- $\int_0^\infty \frac{x^{\eta-1}}{1+x^2} dx$: put $t = x^2$, $dx = \frac{dt}{2\sqrt{t}}$, giving $\frac{1}{2} \int_0^\infty \frac{t^{\eta/2-1}}{1+t} dt = \frac{1}{2} \Gamma(\frac{\eta}{2}) \Gamma(1 - \frac{\eta}{2})$ by (A3).

Expand both for small η , using:

- $\Gamma(\frac{\eta}{2}) = \frac{\Gamma(1+\eta/2)}{\eta/2} = \frac{2}{\eta} - \gamma_E + O(\eta)$;
- $\Gamma(1 \mp \frac{\eta}{2}) = 1 \pm \frac{\gamma_E \eta}{2}$;
- $2^{\eta-1} = \frac{1}{2}(1 + \eta \log 2)$;
- $a^{-\eta} = 1 - \eta \log a$.

Then

$$\text{first} = \frac{1}{2}(1 + \eta \log 2)(1 - \eta \log a) \left(\frac{2}{\eta} - \gamma_E \right) \left(1 - \frac{\gamma_E \eta}{2} \right) = \frac{1}{\eta} + \log 2 - \log a - \gamma_E + O(\eta),$$

$$\text{second} = \frac{1}{2} \left(\frac{2}{\eta} - \gamma_E \right) \left(1 + \frac{\gamma_E \eta}{2} \right) = \frac{1}{\eta} + O(\eta).$$

The $\frac{1}{\eta}$ cancels in the difference:

$$R(a) = \log \frac{2}{a} - \gamma_E = -\frac{1}{2} \log \frac{a^2 e^{2\gamma_E}}{4}.$$

Check by direct integration at $a = 0.3, 1, 3$: 1.3199043, 0.1159315, -0.9826808 , exactly as the formula gives.

$$K_I^{\text{fin}} = -\pi \log \frac{q^2 s^2 e^{2\gamma_E}}{4} = -\pi \log \frac{\xi^2 k^2 (y-x)^2 e^{2\gamma_E}}{4}.$$

Remark. $K_I = K_I^{\text{UV}} + K_I^{\text{fin}} = \pi \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\overline{\text{MS}}}^2}{q^2} \right]$: the $(y-x)^2$ cancels between the two. This is the "momentum form". It is the same number, only split differently.

Step 11. $K_{\text{III}} = \int d^2 r \frac{e^{iq \cdot r}}{(r-s)^2 + M^2}$

(a) Shift the integration variable, $r = s + u$. Then $d^2 r = d^2 u$, $r - s = u$, and $e^{iq \cdot r} = e^{iq \cdot s} e^{iq \cdot u}$:

$$K_{\text{III}} = e^{iq \cdot s} \int d^2 u \frac{e^{iq \cdot u}}{u^2 + M^2}.$$

(b) This is exactly (A4) with $m = M$:

$$K_{\text{III}} = 2\pi e^{iq \cdot s} K_0(|q|M), \quad |q|M = \bar{\xi}|k| \cdot |s| \sqrt{\frac{\xi}{\xi}} = |k||s| \sqrt{\xi \bar{\xi}}.$$

Step 12. $K_{\text{II}} = \int d^2 r e^{iq \cdot r} \frac{2r \cdot s}{r^2 D}$

(a) **Feynman's trick** (A6) with $a = D$, $b = r^2$: $\frac{1}{r^2 D} = \int_0^1 \frac{dt}{[tD + (1-t)r^2]^2}$.

(b) **Simplify the denominator.** Use $D = r^2 - 2r \cdot s + D_0$ (Step 7): $tD + (1-t)r^2 = r^2 - 2tr \cdot s + tD_0$. Complete the square, $r^2 - 2tr \cdot s = (r - ts)^2 - t^2 s^2$:

$$tD + (1-t)r^2 = (r - ts)^2 + \Delta, \quad \Delta \equiv tD_0 - t^2 s^2 = ts^2 \left(\frac{1}{\bar{\xi}} - t \right) > 0.$$

($\Delta > 0$ because $\frac{1}{\bar{\xi}} > 1 \geq t$.)

(c) **Shift** $r = ts + u$. Then $e^{iq \cdot r} = e^{itq \cdot s} e^{iq \cdot u}$ and $2r \cdot s = 2u \cdot s + 2ts^2$:

$$K_{\text{II}} = \int_0^1 dt e^{itq \cdot s} \left[2s_j F^j + 2ts^2 F_0 \right], \quad F_0 = \int d^2 u \frac{e^{iq \cdot u}}{(u^2 + \Delta)^2}, \quad F^j = \int d^2 u \frac{e^{iq \cdot u} u^j}{(u^2 + \Delta)^2}.$$

(d) F_0 . (A4) with $m^2 = \Delta$ gives $\int d^2 u \frac{e^{iq \cdot u}}{u^2 + \Delta} = 2\pi K_0(|q|\sqrt{\Delta})$. Differentiate both sides with respect to Δ (A5).

- Left side: $-F_0$.
- Right side, by the chain rule: $2\pi K'_0(|q|\sqrt{\Delta}) \cdot \frac{|q|}{2\sqrt{\Delta}} = -\frac{\pi|q|}{\sqrt{\Delta}} K_1(|q|\sqrt{\Delta})$, using $K'_0 = -K_1$.

So $F_0 = \frac{\pi|q|}{\sqrt{\Delta}} K_1(|q|\sqrt{\Delta})$.

(e) F^j . Since $\frac{\partial}{\partial u^j} \frac{1}{u^2 + \Delta} = -\frac{2u^j}{(u^2 + \Delta)^2}$, we have $\frac{u^j}{(u^2 + \Delta)^2} = -\frac{1}{2} \frac{\partial}{\partial u^j} \frac{1}{u^2 + \Delta}$. By (A5): $F^j = -\frac{1}{2} (-iq^j) 2\pi K_0(|q|\sqrt{\Delta}) = i\pi q^j K_0(|q|\sqrt{\Delta})$.

(f) **Put together** ($2s_j F^j = 2i\pi(q \cdot s)K_0$):

$$K_{\text{II}} = \pi \int_0^1 dt e^{itq \cdot s} \left[2i(q \cdot s) K_0(|q|\sqrt{\Delta}) + \frac{2ts^2|q|}{\sqrt{\Delta}} K_1(|q|\sqrt{\Delta}) \right], \quad \Delta = ts^2 \left(\frac{1}{\bar{\xi}} - t \right).$$

This is an ordinary one-dimensional integral over t of standard functions.

Step 13. Assemble K ; split it into UV and finite

$K = K_{\text{I}}^{\text{UV}} + K_{\text{I}}^{\text{fin}} + K_{\text{II}} - K_{\text{III}}$. Divide by π :

$$\begin{aligned} \frac{K(\xi)}{\pi} &= \underbrace{\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y-x)^2 e^{2\gamma_E}}{4}}_{\text{UV (Step 9)}} + G(\xi), \\ G(\xi) &= \underbrace{-\log \frac{\bar{\xi}^2 k^2 s^2 e^{2\gamma_E}}{4}}_{\text{Step 10}} + \underbrace{a(\xi)}_{K_{\text{II}}/\pi} - \underbrace{2e^{i\bar{\xi}k \cdot s} K_0(|k||s|\sqrt{\xi \bar{\xi}})}_{-K_{\text{III}}/\pi}, \\ a(\xi) &= \int_0^1 dt e^{it\bar{\xi}k \cdot s} \left[2i\bar{\xi}(k \cdot s) K_0(\bar{\xi}|k|\sqrt{\Delta}) + \frac{2t\bar{\xi}|k|s^2}{\sqrt{\Delta}} K_1(\bar{\xi}|k|\sqrt{\Delta}) \right]. \end{aligned}$$

Check ($s = (0.7, -0.4)$, $k = (1.3, 0.5)$). $\pi G(\xi)$ must equal the directly integrated $\int d^2 r \frac{1}{r^2} \left[e^{iq \cdot r} \frac{s^2}{\xi D} - \frac{s^2}{r^2 + s^2} \right]$. At $\xi = 0.3$: $2.52869189 + 2.18595267i$ direct vs $2.52869143 + 2.18595268i$ formula. At $\xi = 0.6$: $3.12302159 + 0.97391230i$ vs

$$3.12302120 + 0.97391230i.$$

Step 14. Put K back into the cross section

$$\frac{d^3 N_{2b}^{\text{UV}}}{d^3 k} = \mathcal{N} \int e^{-ik \cdot w'} e^{ik \cdot x} \mathcal{W} \mathcal{U}_b \int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}}(\xi) \left\{ \frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y-x)^2 e^{2\gamma_E}}{4} + G(\xi) \right\}.$$

What remains is the ξ (p^+) integral.

Step 15. ξ integral of the UV term

The UV bracket does not depend on ξ , so it multiplies $\int_{\lambda}^{1-\lambda} C_{\text{UV}} d\xi$. Term by term:

- $\int_{\lambda}^{1-\lambda} \xi \bar{\xi} d\xi = \left[\frac{\xi^2}{2} - \frac{\xi^3}{3} \right]_{\lambda}^{1-\lambda} \rightarrow \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.
- $\frac{\xi}{\bar{\xi}} = \frac{1-\xi}{\xi} = \frac{1}{\xi} - 1$, and $\int_{\lambda}^{1-\lambda} \frac{d\xi}{1-\xi} = \left[-\log(1-\xi) \right]_{\lambda}^{1-\lambda} = \log \frac{1-\lambda}{\lambda} \rightarrow L$. So $\int \frac{\xi}{\bar{\xi}} \rightarrow L - 1$.
- In the same way, $\frac{\bar{\xi}}{\xi} = \frac{1}{\xi} - 1$ gives $L - 1$.

$$\int_{\lambda}^{1-\lambda} C_{\text{UV}} d\xi = \frac{1}{6} + 2(L - 1) = 2L - \frac{11}{6}.$$

The UV term of the cross section is therefore $\mathcal{W} \left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right]$.

Step 16. How G behaves as $\xi \rightarrow 1$ (needed for A8)

Write $\ell \equiv \log \frac{k^2 s^2 e^{2\gamma_E}}{4}$ and let $\bar{\xi} \rightarrow 0$. Take the three terms of G one at a time.

- $-\log \frac{\bar{\xi}^2 k^2 s^2 e^{2\gamma_E}}{4} = -\log \bar{\xi}^2 - \ell = 2 \log \frac{1}{\bar{\xi}} - \ell$.
- $z = |k||s| \sqrt{\xi \bar{\xi}} \rightarrow 0$, so by A4 $2K_0(z) = -\log \frac{z^2 e^{2\gamma_E}}{4} = -\ell - \log \xi - \log \bar{\xi} \rightarrow \log \frac{1}{\bar{\xi}} - \ell$. Also $e^{i\bar{\xi}k \cdot s} \rightarrow 1$. So $-2e^{i\bar{\xi}k \cdot s} K_0 \rightarrow -\log \frac{1}{\bar{\xi}} + \ell$.
- $a(\xi) \rightarrow 0$:
 - the K_1 term is $\frac{2ts^2}{\Delta} [z' K_1(z')] \leq \frac{2ts^2}{\Delta} = \frac{2}{1/\xi - t} \leq \frac{2\bar{\xi}}{\xi} \rightarrow 0$ (using $z' K_1(z') \leq 1$);
 - the K_0 term has a factor $\bar{\xi}$ that beats the slowly growing K_0 .

Adding up: $2 \log \frac{1}{\bar{\xi}} - \ell - \log \frac{1}{\bar{\xi}} + \ell$, i.e.

$$G(\xi) \xrightarrow{\xi \rightarrow 1} \log \frac{1}{\bar{\xi}} \quad (\text{no constant term}).$$

$\bar{\xi}$	G	$\log \frac{1}{\bar{\xi}}$
10^{-2}	$4.59435 + 0.00707i$	4.60517
10^{-4}	$9.21009 + 0.00007i$	9.21034
10^{-6}	$13.815507 + 0.000001i$	13.815511

Step 17. How G behaves as $\xi \rightarrow 0$

- $\bar{\xi} \rightarrow 1$, so $-\log \frac{\bar{\xi}^2 k^2 s^2 e^{2\gamma_E}}{4} \rightarrow -\ell$.
- $z = |k||s| \sqrt{\xi \bar{\xi}} \rightarrow 0$, so $2K_0(z) \rightarrow -\log \frac{k^2 s^2 \xi e^{2\gamma_E}}{4} = \log \frac{1}{\xi} - \ell$. So $-2e^{i\bar{\xi}k \cdot s} K_0 \rightarrow -e^{ik \cdot s} (\log \frac{1}{\xi} - \ell)$.
- In $a(\xi)$: $\Delta = ts^2(1-t+\frac{\xi}{\bar{\xi}}) \approx ts^2(1-t+\xi)$ becomes small near $t=1$. There the K_1 term $\rightarrow \frac{2ts^2}{\Delta} \rightarrow \frac{2}{1-t+\xi}$ (using $zK_1(z) \rightarrow 1$), and $\int_0^1 \frac{2dt}{1-t+\xi} = 2 \log \frac{1+\xi}{\xi} \approx 2 \log \frac{1}{\xi}$. Subtracting this piece and adding it back:

$$a(\xi) \rightarrow 2e^{ik \cdot s} \log \frac{1}{\xi} + a_{\text{reg}}, \quad a_{\text{reg}} = \int_0^1 dt \left\{ e^{itk \cdot s} \left[2i(k \cdot s) K_0(|k| \sqrt{\Delta_0}) + \frac{2t|k|s^2}{\sqrt{\Delta_0}} K_1(|k| \sqrt{\Delta_0}) \right] - \frac{2e^{ik \cdot s}}{1-t} \right\},$$

with $\Delta_0 = t(1-t)s^2$. The braces are finite at $t=1$.

Adding up: $-\ell + 2e^{ik \cdot s} \log \frac{1}{\xi} + a_{\text{reg}} - e^{ik \cdot s} \log \frac{1}{\xi} + e^{ik \cdot s} \ell$, i.e.

$$G(\xi) \xrightarrow{\xi \rightarrow 0} e^{ik \cdot s} \log \frac{1}{\xi} + c_0, \quad c_0 = a_{\text{reg}} - (1 - e^{ik \cdot s}) \ell.$$

Test case: $e^{ik \cdot s} = 0.75836 + 0.65183i$, $c_0 = -0.66343 - 0.24594i$.

ξ	G	$e^{ik \cdot s} \log \frac{1}{\xi} + c_0$
10^{-2}	$2.85266 + 2.76828i$	$2.82896 + 2.75587i$
10^{-4}	$6.32169 + 5.75789i$	$6.32134 + 5.75767i$
10^{-6}	$9.813731 + 8.759483i$	$9.813727 + 8.759480i$

Where the $\log \frac{1}{\xi}$ comes from. At $\xi = 0$, $M = 0$ and $\frac{1}{D} = \frac{1}{(r-s)^2}$ is itself log-divergent at $r = s$ (A2). For small ξ the mass M cuts it off, which leaves $\log \frac{1}{M^2} \sim \log \frac{1}{\xi}$. See Step 20 for why this matters.

Step 18. ξ integral of the finite terms (plus prescription)

Near the endpoints:

- $\xi \rightarrow 0$: $C_{\text{UV}} = \frac{\bar{\xi}}{\xi} + \xi \bar{\xi} + \frac{\xi}{\xi} = \frac{1}{\xi} - 1 + O(\xi)$. So $C_{\text{UV}} G = \frac{e^{ik \cdot s} \log \frac{1}{\xi} + c_0}{\xi} + (\text{integrable})$.
- $\xi \rightarrow 1$: $C_{\text{UV}} = \frac{1}{\xi} + O(1)$, so $C_{\text{UV}} G = \frac{\log \frac{1}{\xi}}{\xi} + (\text{integrable})$.

Apply (A8) at both ends, with $A = e^{ik \cdot s}$, $B = c_0$ at $\xi \rightarrow 0$, and $A = 1$, $B = 0$ at $\xi \rightarrow 1$:

$$\int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}}(\xi) G(\xi) = \frac{1 + e^{ik \cdot s}}{2} L^2 + c_0 L + H,$$

$$H = \int_0^1 d\xi \left[C_{\text{UV}}(\xi) G(\xi) - \frac{e^{ik \cdot s} \log \frac{1}{\xi} + c_0}{\xi} - \frac{\log \frac{1}{\xi}}{\xi} \right] \quad (\text{finite}).$$

Check (test case, $H = 0.12277 + 0.36849i$):

λ	direct $\int_{\lambda}^{1-\lambda} C_{\text{UV}} G$	formula
10^{-3}	$37.50412 + 14.22366i$	$37.49192 + 14.22142i$
10^{-4}	$68.59525 + 25.75127i$	$68.59361 + 25.75098i$
10^{-5}	$109.01816 + 40.73650i$	$109.01795 + 40.73649i$

The difference shrinks as $\lambda \rightarrow 0$, as (A8) says.

Step 19. Final answer

Add Step 15 and Step 18 inside Step 14:

$$\frac{d^3 N_{2b\text{NLO}}^{\text{UV}}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2}$$

$$\times \left\{ \underbrace{\left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\mu_{\text{MS}}^2 (y - x)^2 e^{2\gamma_E}}{4} \right]}_{\text{UV term}} + \underbrace{\frac{1 + e^{ik \cdot (y-x)}}{2} L^2 + c_0 L + H}_{\text{finite terms}} \right\}$$

$$\times \left[U^{ab'}(x') - U^{ab'}(w') \right] \left[-N_c U^{af}(x) + N_c U^{af}(y) \right] \rho^{b'}(x') \rho^f(y)$$

- $L = \log \frac{k^+}{\Lambda}$.
- c_0 is given in Step 17 and H in Step 18. Both are finite, dimensionless, and depend only on $k \cdot (y - x)$ and $|k||y - x|$.

- If you want the log in momentum space instead: $\log \frac{\mu^2(y-x)^2 e^{2\gamma_E}}{4} = \log \frac{\mu^2}{k^2} + \ell$, and the ℓ piece moves to the finite terms.

Step 20. One thing you must know before adding the rest of 2b

The term $\frac{1}{2}e^{ik \cdot (y-x)} L^2$ comes only from Step 17, i.e. from $z \rightarrow y$ at $p^+ \rightarrow 0$. It is **not** present in the full 2b:

- In your expression, the $\frac{\bar{\xi}}{\xi}$ term is $\frac{\bar{\xi}}{\xi} \frac{P \cdot s}{P^2 r^2}$, which does not vanish at $r = s$.
- In the full 2b, the $\frac{\bar{\xi}}{\xi}$ block is $T_4 = \frac{\bar{\xi}}{\xi} \frac{P \cdot (s-r)}{P^2 r^2} \left[1 - \frac{s \cdot r}{2s^2}\right]$. It is proportional to $P \cdot (s - r)$ and **vanishes at $r = s$** (checked with sympy).
- Hence the rest of 2b (full 2b minus this piece) contains exactly $-\frac{1}{2}e^{ik \cdot (y-x)} \mathcal{W} L^2$, and the two cancel.

double logs	from $p^+ \rightarrow 0$	from $p^+ \rightarrow k^+$
this UV piece	$+\frac{1}{2}e^{ik \cdot s} \mathcal{W} L^2$	$+\frac{1}{2} \mathcal{W} L^2$
rest of 2b	$-\frac{1}{2}e^{ik \cdot s} \mathcal{W} L^2$	0
full 2b	0	$\frac{1}{2} \mathcal{W} L^2$

So keep this term and let it cancel when you add the rest of 2b. Do not throw it away and do not "renormalise" it: it is not a UV divergence. The UV term of Step 15 is the only UV divergence of 2b.

Numerical checks:

- `src/rowV_tools_check.py` (every toolbox formula A2–A8, and $R(a)$), output in `RESULTS_rowV_tools_check.txt`;
- `src/rowV_uvpiece.py` (Steps 7, 13, 16, 17, 18, 20), output in `RESULTS_rowV_uvpiece.txt`;
- `src/rowV_coordUV.py` (Step 9).